Lesson Title	**SECOND TERM – GLOBAL INTERACTIONS: GLOBAL CLIMATIC PHENOMENA (EL NIÑO SOUTHERN OSCILLATION)**			
Learning Area/s	SCIENCE (Earth Science)			
Name of Teacher/s	IAN B. COROCOTO			
Grade Level and Section	10			
Date and Time	Week 9 (Day 1 to Day 4), 45 minutes per session			
No. of Sessions	**DAY 1**	**DAY 2**	**DAY 3**	**DAY 4**
References *(books, websites, toolkits, etc.)*	• DepEd Grade 10 Science BOW, Second Term, Week 9 • NOAA Climate.gov – El Niño & La Niña • PAGASA – ENSO Alert and Warning System • Science 10 LE/LAS	• DepEd Grade 10 Science BOW, Second Term, Week 9 • PAGASA – ENSO advisories • SunStar Cebu – PAGASA definitions of dry condition, dry spell and drought • GMA News – El Niño 2024 impact report	• DepEd Grade 10 Science BOW, Second Term, Week 9 • PAGASA – La Niña advisories • NOAA Climate.gov – El Niño & La Niña • Science 10 LE/LAS	• DepEd Grade 10 Science BOW, Second Term, Week 9 – ENSO Impact Scenario Activity • PAGASA – ENSO Alert and Warning System; heat index categories • Science 10 LE/LAS
Materials *TV, tablet & materials for each activity*	• Teacher-made PPT • Tray, water, straw, ruler, food coloring, paper bits, marker • Diagrams of the normal Pacific	• Teacher-made PPT • "El Niño Chain Reaction" event cards (7 per group), envelopes • Pacific diagrams	• Teacher-made PPT • "Build the La Niña" cards • "Rainfall Detectives" rainfall chart and data	• Teacher-made PPT • "ENSO Impact Scenarios" cards • "Community ENSO Advisory" worksheet • Short quiz
Declaration of AI use *Cite how AI was used in the formulation of the lesson plan. See DO 3 s.2026 Annex A.*	AI was used to refine the wording, alignment, and flow of the lesson while keeping the prescribed format.	AI was used to refine the wording, alignment, and flow of the lesson while keeping the prescribed format.	AI was used to refine the wording, alignment, and flow of the lesson while keeping the prescribed format.	AI was used to refine the wording, alignment, and flow of the lesson while keeping the prescribed format.
Intentions	Meaningful learning experiences are anchored in how we frame them. Start by deciding what you want learners to master by the end of the lesson – keep it clear and simple. Remember: Understanding your learners' evolving context and designing around it help ensure that your lessons connect with and are relevant to them.			
Learning Competency and Curriculum Standards:	• Describe how global climatic phenomena, such as the El Niño Southern Oscillation, may impact weather systems. Focus for the week: Day 1 – the normal Pacific; Day 2 – El Niño; Day 3 – La Niña; Day 4 – impacts of ENSO on weather and daily life.			
Learning Objectives:	Learning Session 1 **Cognitive:** Explain how trade winds influence warm surface water distribution, upwelling, and Walker circulation under normal conditions in the tropical Pacific Ocean. **Psychomotor:** Demonstrate the effects of varying wind strength on surface water movement and water levels through a simple experiment.	Learning Session 2 **Cognitive:** Analyze how weakened trade winds affect sea-surface temperatures, upwelling, and rainfall patterns during El Niño and their impacts on Philippine weather systems. **Psychomotor:** Assemble a cause-and-effect diagram illustrating the sequence of oceanic and atmospheric changes during El Niño. **Affective:** Advocate responsible water conservation and heat safety practices by	Learning Session 3 **Cognitive:** Differentiate La Niña from El Niño in terms of trade wind strength, sea-surface temperatures, upwelling, and rainfall patterns in the tropical Pacific and the Philippines. **Psychomotor:** Construct a visual model illustrating oceanic and atmospheric conditions during La Niña using directional arrows and weather symbols. **Affective:** Display respect and cooperation by actively participating in group model-making	Learning Session 4 **Cognitive:** Evaluate the potential impacts of El Niño and La Niña on Philippine weather systems and daily life using local weather scenarios. **Psychomotor:** Present a graphic organizer illustrating ENSO phases, associated weather changes, community impacts, and preparedness measures. **Affective:** Promote community preparedness by proposing practical responses to ENSO-

	Affective: Exhibit cooperation and responsibility by actively participating in group activities and observing safety precautions.	suggesting measures to reduce the impacts of El Niño.	and appreciating others' contributions during the gallery walk.	related weather disturbances through a public advisory.
Learner Context:	Learners are expected to connect prior Science learning on weather, climate and ocean–atmosphere interaction with the current lesson on ENSO. They generally respond well to visual prompts, hands-on models, group tasks and real-life Philippine examples. Possible barriers include vocabulary load (e.g., thermocline, upwelling, Walker circulation), abstract global processes, and difficulty explaining cause-and-effect chains. Support will include a clear single objective per day, teacher modeling, diagrams, vocabulary prompts, group discussion, and varied response formats.			
Learning Experience	A learning experience is like a thoughtfully designed journey. Each activity and interaction builds towards meaningful understanding and growth. Identify activities and interactions to help learners gain knowledge, skills, or understanding in a purposeful way.			
Pre-Lesson:	• Classroom routines: prayer, greetings, attendance. • Show headlines about heat, water shortage, floods and crop damage in the Philippines. • Ask: "Why can the same country be too dry one year and too wet the next?" • Present the day's objective.	• Review the normal Pacific with a quick-fire game (trade winds, warm pool, upwelling). • Show a news flash on El Niño 2023–2024: about 40 provinces facing drought, farm damage and a state of calamity in one town. • Ask: "What could be the cause that affects many provinces at once?" • Present the day's objective.	• Review El Niño using a quick recall. • Show photos and headlines of floods, heavy rain and storms during a wet year. • Ask: "Could the Pacific Ocean, thousands of kilometers away, play a part?" • Present the day's objective.	• Review El Niño and La Niña with a short recall. • Introduce the role: learners act as PAGASA analysts and community advisers. • Ask: "How can we tell what is happening in the Pacific from what we see on the ground?" • Present the day's objective.
Flow:	• **Activity: "Wind Pushes Water."** Groups blow across a tray of water to model trade winds; mark the water level at both ends; blow harder and record. • Answer guide questions: tray = Pacific; wind = trade winds; pile-up = warm pool. • Discuss the normal Pacific from above and from the side: trade winds blow east to west, warm water piles up near Asia, cold water upwells near South America. • Explain the Walker circulation: warm moist air rises over the warm pool and brings rain; dry air sinks in the east. • Summarize with "Complete the Thought."	• **Activity: "El Niño Chain Reaction."** Groups arrange 7 event cards in order and add one real-life example of the last event. • Report: weakened trade winds → warm water flows east → upwelling weakens → rain area shifts east → less rain over the Philippines. • Compare normal and El Niño diagrams side by side. • Discuss PAGASA terms: dry condition, dry spell, drought. • Discuss effects on water, farms, heat and the sea (coral bleaching, fish catch). • Summarize with "Complete the Thought."	• **Activity: "Build the La Niña."** Groups assemble the La Niña diagram: stronger trade winds, a bigger warm pool in the west, stronger upwelling. • **Activity: "Rainfall Detectives."** Groups read a rainfall chart and decide which years look like El Niño, La Niña or normal. • Discuss effects of La Niña: above-normal rainfall, floods, landslide risk and more tropical cyclones affecting the Philippines. • Compare El Niño, La Niña and normal in a table. • Summarize with "Complete the Thought."	• **Activity: "ENSO Impact Scenarios."** Groups receive scenario cards from different parts of the Philippines (water shortage, flooding, heat, fish catch, crops). • Groups decide the likely ENSO phase, explain the evidence, and list effects on weather and daily life. • Groups write a **"Community ENSO Advisory"** with 3 safety or preparedness tips and present it. • Teacher processes answers and links them to PAGASA advisories. • Synthesize the whole week.
Learning Resources:	• Science 10 LE/LAS • Teacher-made PPT (Day 1) • Activity sheet and tray set-up • Normal Pacific diagrams	• Science 10 LE/LAS • Teacher-made PPT (Day 2) • Event cards and envelopes • PAGASA terms table	• Science 10 LE/LAS • Teacher-made PPT (Day 3) • La Niña cards • Rainfall chart and data	• Science 10 LE/LAS • Teacher-made PPT (Day 4) • Scenario cards • Advisory worksheet and quiz
Opportunities for integration:	• Mathematics: reading rainfall and sea-temperature values. • Araling Panlipunan: why weather matters to communities.	• Agriculture / TLE: effects of drought on crops and livestock. • Health: heat safety and clean water.	• Disaster risk reduction: flood and landslide preparedness. • English: reading news reports.	• English / Science writing: justify answers using evidence. • Values: cooperation, preparedness and care for the community.
Assessment	Assessments reveal what learners have gained and what they still need help with. These are helpful in providing you with information to guide your future instruction.			
Formative Assessment:	• 5-item multiple-choice quiz on the normal Pacific. • Check the activity sheet and guide questions.	• 5-item multiple-choice quiz on El Niño. • Check the correct order of the chain-reaction cards.	• 5-item multiple-choice quiz on La Niña and the comparison. • Check the rainfall-detective answers.	• Short quiz on ENSO and its impacts. • Check the Community ENSO Advisory using a simple rubric.

Ways Forward	Meaningful learning can also happen beyond the classroom – for both the learners and the teacher. Pause and reflect on what happened today.			
Extended learning opportunities:	• Read about PAGASA's ENSO Alert and Warning System and bring one question about it.	• Search for one news story about drought or a dry spell. Write the date, the place and the effect on people.	• Search for one news story about floods, heavy rain or a La Niña advisory. Write the date, the place and the effect on people.	• Make a one-page family preparedness tip sheet for El Niño or La Niña.
Reflections:				

Prepared by:

IAN B. COROCOTO
Pre-service Teacher

Reviewed by:

MERLIN R. FAVILA
Cooperating Teacher

LYDIA H. PASICOLAN
Head Teacher III, Science Department

Noted:

NOLITO R. ESTILLES
Principal II

Date: _____